

# Collins' projection, verified

The cylindrical algebraic decomposition of `pvs_cad`, built from Collins' projection: the theorem that the projections form a CAD, the decision procedure and the quantifier elimination that read truth values off it, what they cost on the benchmarks, and where the limits are.

**In one line.** For every family of polynomials in any number  $n$  of variables, the cells built from Collins' projection are **proved** to form a cylindrical algebraic decomposition adapted to the family — with no certificate checked at run time — and every cell is defined without quantifiers by sign conditions on polynomials with rational coefficients, so it is semi-algebraic in the sense of Basu–Pollack–Roy. An executable function returns that CAD as data: at least one record, each with an exact sample point, for every nonempty cell of  $\mathbb{R}^n$ , and none for any other. The decision procedure built on it (`decide8`) is **proved** sound and complete, and its quantifier elimination (`qe8`) is **proved** to return a quantifier-free equivalent of every prenex formula with at least one quantifier and one or more free variables; `qelim`, built on both, does the same for every first-order formula. (`cad`) and (`cad-qe`) use them in a theory that imports `pvs_cad`.

## What is proved

### `col_stack.collins_stack` (delineability from Collins' projection)

```
collins_stack: THEOREM conn?(C) AND C(p0) AND colh?(G, C) IMPLIES delin_cl?(G, C)
```

If  $C$  is connected and nonempty and Collins' N-reducta projection of the family  $G$  is sign-invariant on  $C$  (`colh?`: the leading coefficients of the kept reducta, the principal subresultant coefficients  $\text{psc}_j(f, f')$ , and  $\text{psc}_j(f, g)$  for pairs each have one sign on  $C$ ), then  $G$  is *delineable* over  $C$  in the sign-invariant sense (`delin_cl?`; root multiplicities are not tracked): over each point of  $C$ , the real roots of the members not identically zero on its fibre are the values of a constant number of continuous, strictly ordered functions on  $C$ , and every member has one sign on each section and band between them. This is Collins' delineability theorem, the basis of Collins' original algorithm (the smaller projections of McCallum, Brown and Lazard rest on different theorems); here it is proved from the definitions (subresultant theorem, continuity of the roots of a polynomial in its coefficients, connectedness).

### `col_run.col_cad_run` (the projections form a CAD)

```
col_cad_run: THEOREM k >= 1 IMPLIES
  (FORALL p: length(p) = k + 1 IMPLIES
    EXISTS s, A: ... ccell(k, F)(s, A)(p))                % cover
  AND (... ccell(k, F)(s1, A1)(p) AND ccell(k, F)(s2, A2)(p)
    IMPLIES s1 = s2 AND A1 = A2)                          % disjoint
  AND (... conn?(ccell(k, F)(s, A))
    AND fod?(1 + length(A))(ccell(k, F)(s, A)))           % connected, definable
  AND (FORALL s: member(s, csects(k, F))
    IMPLIES cad_over?(ctw(F, k), scell(s)))               % delineable
  AND (... the graph of every root function is definable ...)
```

```

AND (... ccell(k, F)(s, A)(p1) AND ccell(k, F)(s, A)(p2)
      IMPLIES svec(F, p1) = svec(F, p2))                                % F sign-invariant

```

For every family  $F$  in  $k + 1$  variables: project  $k$  times,  $P_k = F$  and  $P_{j-1} = \text{proj}(P_j)$  ( $\text{ctw}(F, k)$  is the tower  $P_1, \dots, P_k$ ), cut the line of the outermost variable at the real roots of the  $k$ -th projection  $P_0$  ( $\text{pbot}, \text{csects}$ ), and build the stacks. The cells of  $\mathbb{R}^{k+1}$  cover it and are disjoint; the cells of every level are connected and first-order definable; every level is delineable over every cell below it, and the graphs of its root functions are definable; and every polynomial of  $F$  has one sign on every cell of  $\mathbb{R}^{k+1}$ . Nothing is checked at run time: the statement has no hypothesis but  $k \geq 1$ , and  $\text{col\_cells}$  adds  $k = 0$ , one variable, where the cells are the sectors of the roots of  $F$ .  $\text{col\_verified}$  adds the data: when the walk's certificate holds (two or more variables), the run returns at least one record for every nonempty cell of  $\mathbb{R}^{k+1}$  and none for any other, each with a sample point in its cell and  $F$ 's signs there;  $\text{col\_found}$  raises the run's effort until the certificate holds, and  $\text{col\_found\_cad}$  says it always returns the CAD as data, in every number of variables, one included. These are the conditions of Basu–Pollack–Roy's Def. 5.1 for a CAD adapted to  $F$ , with the partition stated for  $\mathbb{R}^{k+1}$  (at lower levels it follows from the stack construction,  $\text{tcell\_cover}$  and  $\text{tcell\_disj}$ , and is not restated) and its finiteness given by the records. First-order definable sets (formulas with rational coefficients, no parameters) are exactly the quantifier-free definable ones ( $\text{fod\_qfd}$ , Tarski–Seidenberg, proved for every first-order formula), so every cell is semi-algebraic in the sense of Basu–Pollack–Roy; indeed it is defined by a Boolean combination of sign conditions on polynomials with rational coefficients ( $\text{col\_cells\_sa}$ ), where their definition allows real coefficients.

| Theorem                      | What it says                                                                                                                                                                                               |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\text{ctw\_cad}$            | over any connected set on which the $k$ -th projection is sign-invariant, the tower of projections is a CAD                                                                                                |
| $\text{col\_cad\_run}$       | the projections form a CAD of $\mathbb{R}^{k+1}$ adapted to $F$ , for every $F$ and $k \geq 1$                                                                                                             |
| $\text{col\_cells}$          | the same for every $k$ , one variable included                                                                                                                                                             |
| $\text{col\_verified}$       | under the walk's certificate, the run's records name that CAD's cells of $\mathbb{R}^{k+1}$ , at least one per nonempty cell, with a sample and $F$ 's signs in each                                       |
| $\text{col\_found\_cad}$     | $\text{col\_found}$ , with the search for the run's effort built in, always returns the CAD as data                                                                                                        |
| $\text{col\_found\_decides}$ | its records decide every sentence over $F$ in the same variables, in the same order                                                                                                                        |
| $\text{fod\_qfd}$            | a set is first-order definable exactly when it is quantifier-free definable (Tarski–Seidenberg)                                                                                                            |
| $\text{col\_cells\_sa}$      | every cell at every level, and every root graph, is quantifier-free definable over $\mathbb{Q}$ , hence semi-algebraic                                                                                     |
| $\text{decc\_correct}$       | the decision over the Collins tower, when certified, is the truth of the sentence                                                                                                                          |
| $\text{decc\_complete}$      | for at least two quantifiers some run is certified                                                                                                                                                         |
| $\text{decide8\_decides}$    | $\text{decide8}$ decides every closed prenex formula: its value is the truth (sound and complete)                                                                                                          |
| $\text{qe8\_complete}$       | quantifier elimination by the Collins run: a quantifier-free equivalent of every prenex formula with at least one quantifier and $m \geq 1$ free variables                                                 |
| $\text{qelim\_ok}$           | $\text{qelim}$ (with $\text{qelim\_qf}$ ): a quantifier-free equivalent of every first-order formula, quantifiers anywhere, any $m \geq 0$                                                                 |
| $\text{pstower\_cad}$        | over any connected set on which the projection of its lowest family is sign-invariant, a tower whose families <i>contain</i> the projections of the next is a CAD (used for derivative-closed free levels) |
| $\text{detf\_det}$           | the determinant computed with each minor once ( $n 2^n$ products) is the Laplace determinant ( $n!$ )                                                                                                      |

**What you trust.** The PVS kernel; its ground evaluator, which runs the procedures on the concrete input inside

the proof (proof by reflection); NASALib's saved proofs of the theorems the library imports (from `reals`, `Sturm`, `Tarski`, `structures`, `analysis`, `complex`, `matrices`, `mult_poly` and `interval_arith`, and through them from other NASALib libraries such as `ints`, `power` and `trig`), which were not replayed here; and the reading of the statements (`fsem`, `gsem`, `svec`, `delin_cl?`, `cad_over?` say what is intended). The procedures and the strategy code are not trusted; the procedures' correctness is the theorems.

**Size.** The Collins work is 58 theories with 698 proved formulas (theorems, lemmas and type-correctness conditions) in 3,362 lines of PVS. The whole library has 355 theories and 5,183 proved formulas, all replayed by proveit from a clean copy (2 October 2026, 37 minutes).

## Using it

In a theory, write `IMPORTING pvs_cad (cad/pvs_cad.pvs imports cad_decide8, qe8, gform_ok, mpoly_embed and cad_endgame)`. `(cad)` and `(cad-direct)` then decide a closed formula. A prenex one (one real per quantifier; grouped binders are decided through the nested form) is decided by `decide8`: the Collins run for two or more quantifiers, the sectors of the line for one; the proof instantiates `decide8_correct`. A formula of any other shape (quantifiers inside `NOT`, `AND`, `OR`, `IMPLIES` or `IFF`; binders over `posreal`, `nnreal`, `negreal`, `npreal` or `nzreal`) is decided by `decide_g`, which eliminates its quantifiers from the inside out with `qe8` and `decide8`; the proof instantiates `decide_g_ok`. `(cad :cad-only? t)` answers only through these decisions; with one quantifier that is the sectors of the line, not the Collins run. `(cad-qe FNUM)` eliminates the quantifiers of a formula with free variables by `qe8` (`qe8_correct`), or, for a formula of any other shape, by `qe_g` (`qe_g_ok`). `(cad)` also has shortcuts that `(cad-direct)` and `(cad :cad-only? t)` skip: a witness search, which proves a goal whose quantifiers are all existential (or refutes a hypothesis whose quantifiers are all universal) by one point; with two or more quantifiers, a constant for one variable (existential in a goal, universal in a refuted hypothesis) that settles the matrix outright; and, with three or more, when the full decision does not answer within 10 s, one variable fixed to a constant and the smaller sentence decided. With `cad_decide5` imported instead of `cad_decide8`, the decision is the read-closure `decide5` (`decide7` under `:cad-only?`).

```
c8_l3_quad: LEMMA FORALL (c: real): EXISTS (x1: real): EXISTS (x2: real): EXISTS (x3: real):
  (x1^2 <= 1 AND x2^2 <= 1 AND x3^2 <= 1 AND x1 + x2 + x3 >= c) OR c > 3
Rule? (cad-direct)
Q.E.D.                                [about 5 s; the read-closure decision did not finish in 600 s]
```

`c8_l3_quad` is in `cad_decide8_ex`, which imports `cad_decide8`; the transcript is abridged (the step's messages are left out).

## What it costs

**The decision alone** (`decc_o`, the Collins run, against `decn_o`, the read-closure run, both at  $u = 0$ ; (`cad-mx`) measurements on 1 October 2026, wall clock on one machine; “>” marks a run stopped at that time):

| Formula               | quantifiers | shape                                                      | Collins | read closure |
|-----------------------|-------------|------------------------------------------------------------|---------|--------------|
| <code>t_circle</code> | 2           | $\forall x \exists y. x^2 + y^2 = 1 \vee x^2 > 1$          | 0.1 s   | 0.0 s        |
| <code>t_sphere</code> | 3           | the same in three variables                                | 0.1 s   | 0.2 s        |
| <code>e4_above</code> | 4           | $\forall x y z \exists w. w > x \wedge w > y \wedge w > z$ | 2.1 s   | 0.9 s        |

|                            |     |                                                            |       |        |
|----------------------------|-----|------------------------------------------------------------|-------|--------|
| e4_between                 | 4   | the same with $w < z$ (false)                              | 0.2 s | 0.4 s  |
| k2_lin                     | 3   | box $[-1, 1]^2$ against $x_1 + x_2 \geq c$ , $c$ universal | 0.1 s | 0.2 s  |
| l2_quad                    | 3   | the same with $x_i^2 \leq 1$                               | 0.1 s | 0.8 s  |
| k3_lin                     | 4   | box $[-1, 1]^3$ against $x_1 + x_2 + x_3 \geq c$           | 0.5 s | 11.5 s |
| l3_quad                    | 4   | the same with $x_i^2 \leq 1$                               | 0.7 s | >600 s |
| bath_01                    | 3   | ball and cylinder (Bath CAD bank)                          | 0.2 s | >90 s  |
| bath_08                    | 3   | Random B, Gröbner form (false)                             | 0.8 s | 1.1 s  |
| bath_02-05, 07, 09, 10, 12 | 3-4 | the other Bath problems                                    | >90 s | >90 s  |

**Every proof of the example theories**, replayed through (cad), (cad-direct) or (cad-qe), including the witness search and the reconstruction of the proof: as they are, importing cad\_decide5 (qe\_all for qe\_cad\_ex), and with cad\_decide8 (qe8 for qe\_cad\_ex) added to their imports. Measured on 1 October 2026 with the strategies of that day, before the changes for formulas as people write them:

| Theory                           | proofs | cad_decide5 / qe_all | cad_decide8 / qe8  |
|----------------------------------|--------|----------------------|--------------------|
| cad_examples,<br>cad_examples2-4 | 28     | 69.1 s               | 90.2 s             |
| cad_endgame_ex                   | 18     | 31.0 s               | 43.0 s             |
| cad_star_ex                      | 15     | 39.9 s               | 49.5 s             |
| cad_bath                         | 1      | 9.5 s                | 10.2 s             |
| qe_cad_ex                        | 6      | 42.0 s               | 51.1 s             |
| all                              | 68     | 191.5 s, 68 proved   | 244.0 s, 68 proved |

In that run every proof went through either way, and again in a re-run after the strategy fixes of this release (68 of 68 with each import, 1 October 2026). The whole (cad -1) proof of cad\_bath.bath\_08\_false (Bath 08, “Random B, Gröbner form”, false) is one of them: 9.5 s as it is and 10.2 s with cad\_decide8. On these small examples, in that first run, a proof took 0.77 s longer on average with cad\_decide8 or qe8 imported: 0.1–1.3 s for a (cad) or (cad-direct) proof, 0.5–2.5 s for a (cad-qe) proof (in the re-run after the strategy fixes, 0.59 s on average: 166.7 s against 206.9 s for the 68 proofs). Part of that is the larger imported context, not the Collins run: a sentence with one quantifier runs the same decision, decq2, under either import, and the 16 one-quantifier proofs of cad\_endgame\_ex are 0.2–1.1 s slower (0.64 s on average). On some problems with several quantifiers the Collins run is faster by more than an order of magnitude (k3\_lin: 23 times; bath\_01 and l3\_quad: more than 450 times, and the whole proof of l3\_quad through (cad-direct) takes about 5 s, against more than 600 s for the read-closure decision alone); on others it is slower (e4\_above: 2.1 s against 0.9 s).

## How it works

1. **The projection** (projn): for each polynomial of the family, the leading coefficients of its kept reducta (its coefficients from the top down to the first that is a nonzero constant) and the principal subresultant coefficients  $\text{psc}_j(f, f')$  of each kept reductum  $f$ ; for every pair of polynomials,  $\text{psc}_j(f, g)$  for their kept reducta  $f$  and  $g$ ; normalized, without constants and repetitions. The psc are determinants of Sylvester-type matrices, computed with every minor once (detf).
2. **Delineability** (collins\_stack): over a connected set where the projection is sign-invariant, the number of

distinct roots is constant (equal psc patterns of  $(f, f')$  give equal numbers of distinct roots), the roots move continuously (lower semicontinuity of the roots in the coefficients), one root stays in each small disc, and common roots of two members stay common (the subresultant  $S_k$ ); so the root functions are continuous and never cross.

3. **The tower:** projecting  $k$  times gives  $P_{k-1}, \dots, P_0$  below  $P_k = F$ ; over each sector of the roots of  $P_0$  the stacks of  $P_1, \dots, P_k$  form the CAD (ctw\_cad), because each level's projection is sign-invariant on the cells of the level below.
4. **Truth values** come from the verified walk of the read-closure engine, over exact algebraic sample points: it passes the sections and bands of each level and folds the quantifiers. Its certificate only checks the answers of the sign oracle at the samples; delineability no longer depends on it.

## Limits

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- **The size of Collins' projection.** It contains psc of every pair of polynomials and of every kept reductum. On bath\_04 (one polynomial in three variables, of degree 4 in its main variable) the first projection has 11 polynomials and the second 569; isolating their real roots is where the time goes, and most problems of the Bath bank do not finish in 90 s with either engine. The remedies are known and are not done here: a smaller projection (McCallum's or Brown's; cad\_proj defines a McCallum-shaped operator, not proved delineating; the read-closure engine starts from it, under its run-time certificate), equational constraints, square-free factors. Measured with decide5 (cad\_decide5), September 2026: (cad) settles Bath 01, 03 and 04 by a model search, and Bath 02 — and 05 and 07, refuted as hypotheses — by fixing one variable to a constant and deciding the smaller sentence, after the full decision does not answer within its 10 s budget. With cad\_decide8 imported (IMPORTING pvs\_cad, 1 October 2026) the same six are settled the same way: Bath 01, 03 and 04 in 0.5–0.6 s, Bath 02, 05 and 07 in about 12 s each (whole proofs; the smaller sentences are decided by decide8).
- **Universal sentences with many polynomials** (the Joukowski problems bath\_09, 10 and 12) get no help from a witness search, and their decision does not finish in 90 s with either engine.
- **Quantifier elimination with two or more free variables** first answers by sign conditions on the projection polynomials of the plain tower; when they do not tell the true cells from the false ones (two cells with the same signs and different truth values), qe8 runs the Collins tower whose free families are closed under the derivative (ctwd), where by Thom's lemma they always do. The answer is always quantifier-free; the derivative-closed tower is larger, so such problems cost more.
- **Determinants** are computed with every minor once:  $n 2^n$  products instead of  $n!$ ; a fraction-free elimination would be cubic.
- **Where it is.** The theories are in cad/: the col\_\*, decc\_\*, cad\_decide8\*, qe1c\_\*, qe2cc\_\*, qe2cd\_\*, ctwd\*, twrun\_\*, qe8\* and qelim\_\* files and cad\_sa, with the algebra they rest on (sres\_\*, cpoly\_\*, croot\_\*, ev\_near, rdet\_\*, det\_fast\*, cad\_projn\*). cad/top.pvs lists and describes every theory of the library.